

LATENT PLANNING VIA EMBEDDING ARITHMETIC: A CONTRASTIVE APPROACH TO STRATEGIC REASONING

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ABSTRACT

Can planning be carried out directly in a learned representation, without ever training a policy, a value head, or a dynamics model? This paper answers that question in the affirmative and makes three contributions. *First*, we introduce SOLIS, a method that learns an evaluation-aligned embedding space with supervised contrastive learning, so that outcome similarity is captured by proximity and a single global *advantage vector* orients the space from losing to winning regions. *Second*, we show that this representation reduces planning to vector arithmetic: candidate actions are ranked by their alignment with the advantage direction, so that a shallow search over these scores is sufficient to act. *Third*, we validate the approach in chess, where SOLIS attains competitive strength under constrained conditions while searching far fewer nodes than a conventional engine, and where real games trace smooth, interpretable paths through the embedding space. Together, these results establish evaluation-aligned latent planning as a lightweight alternative to traditional dynamics models or policy learning.

1 INTRODUCTION

Planning in high-dimensional state spaces remains a central challenge in artificial intelligence. Traditional reinforcement learning methods rely on explicit search or learned policies, but these can become inefficient when the space of possible actions grows large (Jiang et al., 2023; Gieselman and Pokorny, 2022). A recent line of work has explored latent planning, which reasons over compact latent representations rather than raw states, thereby reducing complexity and enabling more human-like decision-making (Hafner et al., 2019; 2020; Schrittwieser et al., 2020). These methods, however, still pair the learned representation with an additional learned component: a policy, a value head, or a dynamics model. In this paper we ask a sharper question: can the representation *alone* carry the planning?

Our answer is **SOLIS** (*Strategic Optimization via Latent Interpolative Spaces*), a chess engine that treats decision-making as interpolation between regions of advantage in a learned embedding space. SOLIS is a transformer encoder trained with supervised contrastive learning to embed states into an evaluation-aligned latent space, in which distance correlates with similarity of outcome and directional movement corresponds heuristically to progress toward favorable states. The key idea is that this geometry can be summarized by a single global “advantage vector” pointing from losing to winning regions, so that ranking candidate moves by their alignment with this direction is enough to plan—no policy, value head, or dynamics model required.

Contributions. This paper makes the following contributions:

- **Evaluation-aligned latent space.** We show how supervised contrastive learning, driven by evaluation margins, produces an embedding in which proximity reflects outcome similarity and a single global advantage vector orients the space from losing to winning regions (Section 3).
- **Planning as vector arithmetic.** We reduce action selection to scoring candidate embeddings by their alignment with the advantage vector, and introduce two variants of this rule—unanchored and anchored—so that a shallow search suffices to act (Section 3.4).
- **Competitive strength with far less search.** We show empirically that SOLIS reaches 2500+ Elo with only a 5-ply search under 50 ms/move blitz conditions, and that at equal depth $S \in \{3, 4, 5\}$

it is competitive with a configured Stockfish baseline while searching far fewer nodes per move (Section 4).

- **Interpretable trajectories.** We demonstrate that real games trace smooth, readable paths through the embedding space, giving qualitative insight into how planning unfolds (Section 4).

The unifying theme across these contributions is a single claim: an evaluation-aligned representation is enough to plan. In contrast to latent planning methods that couple learned dynamics with policy or value heads (Hafner et al., 2019; 2020; Schrittwieser et al., 2020), SOLIS relies solely on the representation of an evaluation-aligned space for action selection. Although we use chess as a testbed, this vector-based latent planning approach may extend to other perfect-information domains where state evaluations are available or learnable.

2 BACKGROUND

Planning in very large action spaces presents a formidable challenge in artificial intelligence because the number of potential action sequences grows exponentially with every additional decision step. To make our contributions precise, this section situates SOLIS against the landscape of planning and learning it departs from: we review the formal setting (Markov decision processes), the model-based methods that motivate our design, the chess engines we compare against, and the contrastive-learning tools we build on. Throughout, we highlight the point of departure that defines this work—SOLIS plans in the representation alone, without a learned policy, value head, or dynamics model.

2.1 MARKOV DECISION PROCESSES

Sequential decision making tasks are formalized as Markov Decision Processes (MDPs), a tuple $\{\mathcal{S}, \mathcal{A}, T, R, p(s_0), \gamma\}$ in which $\mathcal{S}$ denotes the state space, $\mathcal{A}$ the action space, T the transition model, R the reward function, $p(s_0)$ the initial state distribution, and $\gamma \in [0, 1]$ the discount factor for future rewards (Puterman, 2014). A policy $\pi : \mathcal{S} \rightarrow p(\mathcal{A})$ maps states to probability distributions over actions. We introduce this vocabulary because a central contribution of SOLIS is precisely that it plans *without* learning such a policy—a point we make concrete once the alternatives are laid out below.

In this setting, a *model* refers to reversible access to the MDP dynamics: given a state–action pair, the model predicts the next state and reward (Moerland et al., 2022). For games such as chess and Go the rules provide a direct model; where the rules are not known, the model must be approximated from data (Gieselsmann and Pokorný, 2022). SOLIS falls in the former case—chess supplies exact dynamics—which lets us isolate our contribution to the representation and the search rule rather than to model learning.

Planning and reinforcement learning are distinguished by how they use this access to the dynamics (Moerland et al., 2022). Planning methods assume access to a model and simulate from the current state to choose an action, much as humans plan in their minds (Russell and Norvig, 1995); because the solution is relative to the current state, it is *local* and is discarded after execution. Reinforcement learning methods, by contrast, typically lack reversible access to the dynamics (Sutton and Barto, 1998). They interact with the environment step by step and consequently learn a *global solution*, such as a policy or value function, that generalizes across the state space. This distinction frames our contribution precisely: SOLIS uses the known model to plan locally, but its global learned object is neither a policy nor a value function—it is the evaluation-aligned embedding itself.

Model-based reinforcement learning combines both elements, using a model while also learning a global solution. Following Moerland et al. (2022), this takes two forms: model-based RL with a *learned* model (e.g., Dyna (Sutton, 1991)), where both the dynamics and the global solution are learned, and model-based RL with a *known* model (e.g., AlphaZero (Silver et al., 2017)), where the dynamics are given and only the global solution is learned. SOLIS is closest to the latter branch, but replaces its learned policy and value heads with a single evaluation-aligned representation.

2.1.1 MODEL-BASED RL WITH A LEARNED MODEL

A large body of work falls into the category of model-based RL with a learned model, where both the dynamics and the global solution are learned. PlaNet (Hafner et al., 2019) learns a latent dynamics

model and combines it with a learned global solution. Dreamer (Hafner et al., 2020) and its successors DreamerV2 (Hafner et al., 2022) and DreamerV3 (Hafner et al., 2024) scale this approach to more complex tasks while still maintaining global value functions and policies. MuZero (Schrittwieser et al., 2020) likewise learns a latent dynamics model, but integrates it with tree search to strengthen planning while training a global policy and value function. Further examples include Dyna (Sutton, 1991) and MBPO (Janner et al., 2021), both of which use learned models to support value or policy learning. What unites this family—and what SOLIS deliberately omits—is the reliance on an explicit learned dynamics model in addition to a policy or value head.

2.1.2 MODEL-BASED RL WITH A KNOWN MODEL

When the environment dynamics are available, the algorithm can focus on coupling search with learning. AlphaZero (Silver et al., 2017) uses exact game rules as the model, combining Monte Carlo tree search (MCTS) with learned policy and value heads. SAVE (Hamrick et al., 2020) similarly assumes a simulator and integrates Q-learning with MCTS: a learned Q-function serves as a prior for search, while search-produced Q-estimates are amortized back into the network. SOLIS shares the “known model” assumption with these engines but differs in what it learns: rather than a policy, a value function, or a Q-function that guides search, it learns only an evaluation-aligned embedding, and the guidance for search comes from geometry within that embedding.

2.1.3 LATENT PLANNING

Latent planning simplifies decision-making by operating in a compressed representation space rather than the raw state space. PlaNet (Hafner et al., 2019), Dreamer (Hafner et al., 2020), and MuZero (Schrittwieser et al., 2020) pair latent dynamics models with value or policy heads, enabling planning through simulations in the learned space. Expansive Latent Space Trees (ELAST) (Giesemann and Pokorny, 2022) cast latent planning as explorative tree search, combining contrastive representations with sampling-based expansion to address long-horizon control. Trajectory Autoencoding Planner (TAP) (Jiang et al., 2023) takes a similar route, learning a compact latent action space and planning over discrete action codes. Notably, all of these methods still simulate or roll out within the latent space, and all attach at least one auxiliary learned head.

This is the gap our first two contributions target. SOLIS trains no policy or value head and learns no dynamics model; instead it distills evaluation targets directly into the geometry of the representation. Planning then reduces to moving between regions aligned with favorable outcomes, using only the representation space at inference—no rollouts, no simulation of futures, only a scoring of candidate embeddings against a learned advantage direction.

2.2 CHESS ENGINES

IBM scaled chess computing in 1997 when its Deep Blue engine defeated the reigning world champion, Garry Kasparov, in a head-to-head match. Early versions of Stockfish (Romstad et al., 2008) followed Deep Blue’s recipe: handcrafted evaluation functions paired with deep alpha-beta tree search. The methodology achieved superhuman performance, but it became clear that the limiting factor was the combined chess mastery of those designing the static evaluation function.

The rating bottlenecks associated with static evaluation inspired a vein of research independent of human guidance. AlphaZero (Silver et al., 2017) and its open-source counterpart Leela Chess Zero (Pascutto and Linscott, 2018) made a significant leap to engines that learned to evaluate positions entirely through self-play reinforcement learning, with no prior knowledge beyond the rules of chess. Modern versions of Stockfish likewise adopted an efficiently updatable neural network evaluation module (NNUE) (Nasu, 2018) to accelerate position evaluation while retaining traditional alpha-beta search.

While these advances produced engines that far surpass human players, the underlying *search* has remained largely unchanged: it still follows principles first proposed by v. Neumann (1928) and Shannon (1950), later integrated into Deep Blue, and relies on deep alpha-beta search through millions of states (Lai, 2015). This is exactly where our third contribution enters. Rather than accelerating evaluation within the same broad search, we ask whether contrastive learning can change the *nature* of the search itself—treating planning as traversal through a high-dimensional embedding

space—so that far fewer nodes need to be examined. Chess thus serves as a demanding, well-instrumented testbed for the general claim that an evaluation-aligned representation can substitute for broad search.

2.3 CONTRASTIVE LEARNING

Contrastive learning is the tool that makes our first contribution possible, so we review it here. It encourages a model to embed similar inputs closer together and dissimilar inputs farther apart in a latent space (Chen et al., 2020). Concretely, given a batch I of training examples containing an anchor input x_i , an encoder $f(\cdot)$ maps the anchor to a normalized embedding $z_i = f(x_i)$. Positive pairs (z_i, z_j) correspond to semantically similar inputs, while all other pairs of samples in the batch are treated as negatives. The design decision that turns this generic machinery into an evaluation-aligned space—defining “similar” by closeness of evaluation—is what we develop in Section 3.

A common contrastive objective is InfoNCE (van den Oord et al., 2019), defined as:

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \in A(i)} \exp(\text{sim}(z_i, z_k)/\tau)}, \quad (1)$$

where $A(i) = I \setminus \{i\}$ is the set of all samples in the batch excluding the anchor i , $\text{sim}(\cdot, \cdot)$ denotes a similarity metric, and τ is a scalar temperature parameter. In the self-supervised case, the positive sample z_j is an augmentation of the anchor z_i and labels are not required (Chen et al., 2020; Khosla et al., 2021).

When labels *are* available—as they are for us, in the form of position evaluations—the supervised contrastive loss (SupCon) (Khosla et al., 2021) extends InfoNCE by allowing multiple positive examples per anchor. This is the objective we adopt, and it is central to our first contribution because it lets us tie “positive” directly to evaluation similarity. The SupCon loss is defined as:

$$\mathcal{L}_{\text{sup}} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\text{sim}(z_i, z_p)/\tau)}{\sum_{a \in A(i)} \exp(\text{sim}(z_i, z_a)/\tau)}, \quad (2)$$

where $P(i)$ denotes the set of positive examples corresponding to anchor i and $A(i)$ is defined as above. Increasing the number of positive and negative pairs has been shown to improve representation quality (Khosla et al., 2021; Chowdhury et al., 2024), a property we exploit when constructing evaluation-margin positives.

3 METHODS

We now describe the components of our approach. The section is organized around our first two contributions: how we *learn* the evaluation-aligned space (tokenization, encoder architecture, and supervised contrastive training) and how we *plan* in it as vector arithmetic (embedding-guided search). Read together, these subsections make concrete the central claim that no policy, value head, or dynamics model is needed—only the representation and a single advantage direction.

3.1 INPUT REPRESENTATION AND TOKENIZATION

Each chess position is represented by its Forsyth-Edwards Notation (FEN) ASCII string, which compactly encodes piece placement, side to move, castling rights, en passant target, half-move clock, and full-move counter. Following the tokenization scheme of Ruoss et al. (2024), we tokenize each FEN into a fixed-length sequence of 77 tokens by expanding run-length encodings. As Ruoss et al. (2024) note, this tokenization renders chess non-Markovian, since information about past states is omitted yet can matter in special cases (e.g., threefold repetition or the fifty-move rule). We adopt this representation deliberately for its simplicity, and we return to the trade-off it introduces—one of the two limitations we analyze—in Section 5.

3.2 ENCODER ARCHITECTURE

Our encoder is a multi-layer transformer (Vaswani et al., 2023) adapted for tokenized chess positions, and it is the only learned object in the entire system—there is no separate policy or value

network. To probe how much of our strength contribution is due to scale, we train two variants, SOLIS Base and SOLIS Mini, whose configurations are summarized in Table 1; the contrast between them recurs throughout the results.

Table 1: Transformer model configurations used in our experiments. The two variants let us separate the strength contributed by the learned space from the contribution of raw model scale (0.8M vs. 41M parameters).

Model	Layers	Embedding dim. D	Heads	MLP Size	Params
SOLIS Mini	6	128	8	256	0.8M
SOLIS Base	6	1024	16	1024	41M

Each input sequence is embedded into D -dimensional vectors through a learned token embedding matrix. A special classification token (CLS) (Devlin et al., 2019; Dosovitskiy et al., 2021) is prepended to aggregate information across the input. Because the input sequences are fixed length, we add a learned positional encoding to each token embedding (Ruoss et al., 2024). The resulting sequence is processed by stacked transformer encoder layers with GELU (Hendrycks and Gimpel, 2023) activations and dropout (Srivastava et al., 2014). The final hidden state at the CLS token is extracted, passed through a linear projection, and ℓ_2 normalized. This normalized CLS embedding is the sole quantity SOLIS reasons over at inference—the single representation that carries all planning in place of a policy or value head.

3.3 SUPERVISED CONTRASTIVE TRAINING

This subsection delivers our first contribution: turning evaluation labels into an evaluation-aligned geometry. We train the encoder on 5 million randomly sampled positions from the ChessBench dataset (Ruoss et al., 2024), each annotated with a Stockfish-evaluated win probability for the player to move. We normalize all evaluations to White’s perspective, so that values near 1.0 indicate a decisive advantage for White and values near 0.0 an advantage for Black.

The crucial design choice is how we define “similar.” We set δ as the evaluation margin for identifying similar positions. For each anchor we precompute all positions whose win probabilities differ by less than δ and randomly sample five as positives during training; to build the batch-level mask, we compare the win probabilities of all samples and mark pairs as positive when their evaluation difference falls below δ . This margin-based sampling pulls together states of nearly equal evaluation and pushes apart positions with larger gaps. The consequence is precisely the property our planning rule later exploits: the embedding space acquires an ordering from losing to winning positions.

We apply supervised contrastive learning (SupCon) (Khosla et al., 2021) with these batch masks to define positive and negative pairs. Pairwise scores use cosine similarity, with temperature $\tau = 0.07$ (Girdhar et al., 2023; Radford et al., 2021) controlling the sharpness of the distribution. Optimization uses stochastic gradient descent with momentum 0.9 (Dosovitskiy et al., 2021; Chen et al., 2020; He et al., 2015), batch size 128, for 400,000 steps, on six NVIDIA L40S GPUs (48 GB each) with data parallelism (Paszke et al., 2019). Complete training hyperparameters appear in Appendix A.2.

Figure 1 makes this contribution visible. We project the Base model’s representation space with UMAP (McInnes et al., 2020); each point is an encoded chess position from a test dataset, colored by its Stockfish-evaluated win probability for White. Positions strongly favoring White appear in gold (*White has checkmate*), and those strongly favoring Black in dark green (*Black has checkmate*). The orderly transition of color across the projection confirms that the training procedure has produced exactly the evaluation-aligned space our planning rule assumes.

3.4 EMBEDDING-GUIDED SEARCH

Here we develop our second contribution: reducing planning to vector arithmetic. At inference SOLIS plans directly in the learned space by ranking legal continuations along a single global advantage direction—no rollouts and no learned policy. We compute means from a held-out dataset at

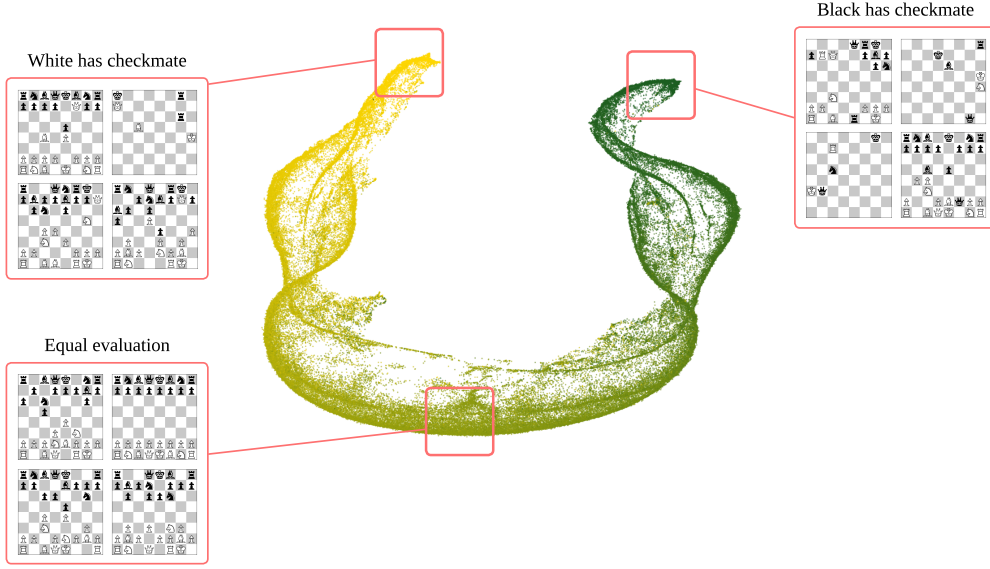


Figure 1: Direct evidence for our first contribution: a UMAP projection of the Base model’s learned embedding space, colored by win probability (gold = White favored, green = Black favored). The smooth color gradient shows that proximity encodes outcome similarity and that the space is oriented from losing to winning regions—the structure the advantage vector summarizes.

the extremes of evaluation,

$$\mu_{\text{White}} = \mathbb{E}[z \mid p=1.0], \quad \mu_{\text{Black}} = \mathbb{E}[z \mid p=0.0],$$

and define the normalized advantage vector

$$\vec{a} = \frac{\mu_{\text{White}} - \mu_{\text{Black}}}{\|\mu_{\text{White}} - \mu_{\text{Black}}\|}.$$

This single vector is the entire planning signal. We consider two mechanisms to convert it into scalar scores, and we evaluate both throughout the results.

3.4.1 UNANCHORED (GLOBAL-DIRECTIONAL)

The first mechanism scores each child embedding directly by its alignment with the advantage direction. Each child embedding z' is scored by

$$\text{score}_{\text{dir}}(z') = \langle z', \vec{a} \rangle,$$

favoring moves whose embeddings extend furthest along the advantage direction regardless of the origin. This is the scoring mechanism illustrated in Figure 2.

3.4.2 ANCHORED (RELATIVE PROJECTION)

The second mechanism shifts by the Black mean and scores

$$\text{score}_{\text{anch}}(z') = \langle z' - \mu_{\text{Black}}, \vec{a} \rangle,$$

so that the value is proportional to signed distance from μ_{Black} along $\vec{a}$. Removing dependence on an arbitrary origin, this variant provides a more stable basis for comparing continuations in practice, and our results confirm that it delivers a small but consistent Elo gain over the unanchored rule.

Crucially, both mechanisms share the same lightweight search (Figure 2). From the “Starting board state” we “Extract possible moves” and embed each child in parallel with the “Transformer encoder”. We then compute either $\text{score}_{\text{dir}}$ or $\text{score}_{\text{anch}}$ in the “Cosine similarity” block and keep the

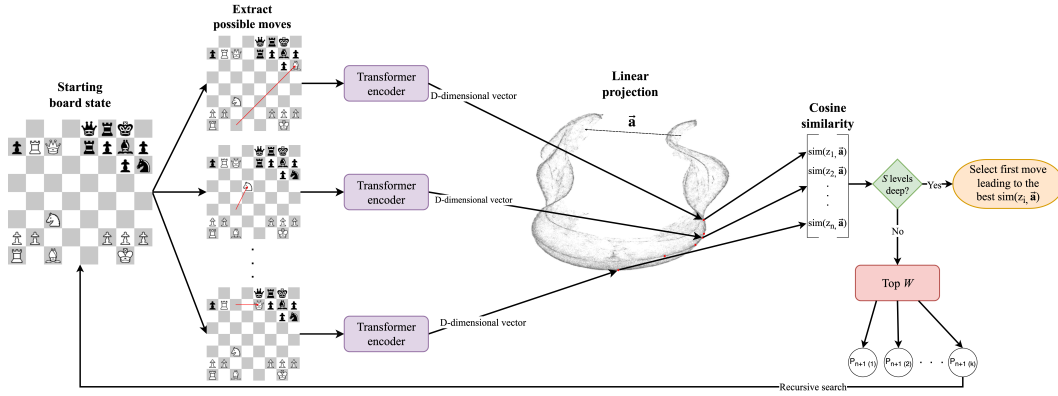


Figure 2: System overview of SOLIS, summarizing our second contribution: planning as vector arithmetic. Candidate moves are embedded and scored by their alignment with the global advantage vector, with no policy, value head, or dynamics model in the loop. For simplicity, the diagram shows the unanchored scoring mechanism $\cos(z', \vec{a})$.

“Top- W ” children. If we have reached “ S levels deep?” the procedure terminates and we “Select first move...”; otherwise we continue the “Recursive search” with min-max preference (White maximizes, Black minimizes). We employ Zobrist hashing (Zobrist, 1990) for a transposition table to avoid re-evaluating positions. The narrow branching factor W and shallow depth S are what make this contribution matter empirically: the search stays tiny because the representation, not broad expansion, does the work.

4 RESULTS

Our experiments are organized to substantiate the paper’s third and fourth contributions. We first establish that SOLIS reaches competitive strength (Elo scaling in Section 4.1), then show that this strength is achieved while searching far fewer nodes than a conventional engine (equal-depth simulations), and finally demonstrate the interpretability of the learned space (game trajectories). Throughout, the recurring takeaway is that the quality of the evaluation-aligned representation—not broad search—drives performance.

4.1 RATING CALCULATION

To estimate Elo ratings, each model was evaluated against Stockfish with a fixed per-move time limit of 50 ms (Ruoss et al., 2024). Full Stockfish configurations are provided in Appendix A.3. To characterize how strength scales, we ablated both the breadth and depth of the search. At each depth/width configuration, our models played 600+ games with alternating colors against Stockfish configured at different Elo caps, and we evaluated both planning rules (unanchored and anchored) under this setup. Caps were selected so that each model configuration yielded at least one matchup with a positive win rate and one with a negative win rate. Elo ratings and 95% confidence intervals were estimated using Bayesian logistic regression via the BayesElo (Coulom, 2008) program with the default confidence parameter of 0.5. Elo estimates for the unanchored rule appear in Figure 3 and for the anchored rule in Figure 4. Stockfish rating caps and detailed match results are reported in Appendix A.6 for reproducibility.

Unanchored ratings Elo scales consistently with search under the unanchored rule, and the pattern directly supports our efficiency claim. Depth provides the largest gains, while width jumps clearly from $W \in \{1, 2\}$ to $W=3$ but improves little beyond, plateauing at $W=5$. This plateau is the key observation: the strongest continuations generally lie within the top three candidates, so expanding further adds almost nothing—strength comes from the space, not the breadth. At shallow depths SOLIS Base leads Mini by roughly 100–250 Elo; at depths 4 and 5 Mini becomes competitive with Base and even overtakes it at $W \in \{3, 5\}$ for this rule.

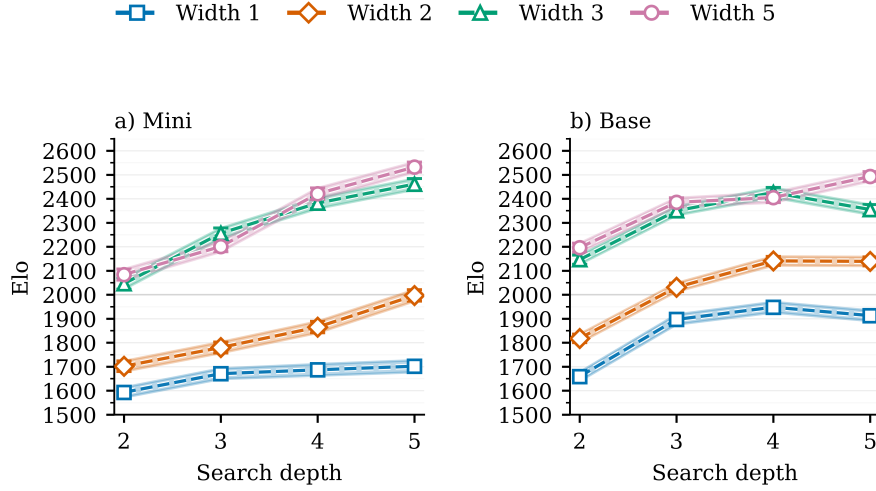


Figure 3: Evidence for our strength contribution: Elo for the Mini and Base models across search widths and depths under the *unanchored* planning rule. Depth drives the gains while width saturates by three candidates, supporting the claim that representation quality—not broad expansion—is the main lever. Shaded bands denote 95% confidence intervals.

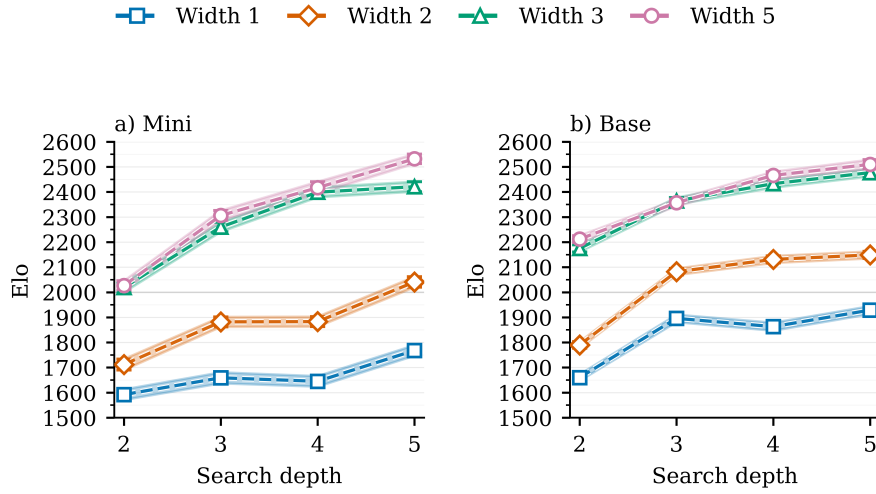


Figure 4: Elo for the Mini and Base models across search widths and depths under the *anchored* planning rule. Both variants exceed 2500 Elo at depth five, and anchoring adds a consistent 10–30 Elo for Base—quantifying the benefit of the second scoring mechanism from our planning contribution. Shaded bands denote 95% confidence intervals.

Anchored ratings With anchored scoring, Elo estimates rise smoothly with both depth and width, and the Base–Mini gap holds in the 100–250 range. Both models exceed 2500 Elo at depth 5, but—unlike the unanchored case—Base is consistently stronger than Mini across all settings. Anchoring further yields a steady 10–30 Elo gain for Base, confirming that shifting relative to the Black mean provides a more reliable basis for comparison and validating the second variant of our planning rule.

4.2 EQUAL-DEPTH SIMULATIONS

This experiment isolates the core of our efficiency contribution: at *equal* search depth, does SOLIS hold its own against a conventional engine? Guided by the Elo scaling above, we fix the branching

factor at $W=3$, since wider searches added only marginal gains and the strongest continuations lie among the top three candidates. Stockfish, by contrast, uses its default expansion with no fixed branching factor. At each depth $S \in \{3, 4, 5\}$ we simulate 200 games with alternating colors; results for both scoring rules appear in Table 2.

Simulation results The head-to-head outcomes back the claim of competitive strength. With unanchored scoring, SOLIS Base outscores Stockfish at every tested depth, while Mini is comparable at depths 3 and 5 and trails only at depth 4. With anchored scoring, Base loses narrowly at depth 3 but outscores Stockfish at depths 4 and 5, and Mini wins at depths 3 and 5 while again falling behind at depth 4. Base is therefore competitive with, and often ahead of, the baseline at matched depth.

Table 2: Equal-depth match points from SOLIS vs. Stockfish, supporting our competitive-strength contribution. The middle portion reports the direct-similarity (unanchored) rule and the rightmost portion the anchored-projection rule; higher is better and draws count as 0.5. Bold marks the higher score in each matchup: SOLIS Base leads in five of six configurations.

Config	Depth	Direct Similarity		Anchored Projection	
		SOLIS score	Stockfish score	SOLIS score	Stockfish score
Base	3	106	94	98	102
	4	109	91	108	92
	5	107.5	92.5	110	90
Mini	3	108	92	105	95
	4	85	115	91	109
	5	101	99	113	87

Nodes searched This is where the efficiency contribution becomes quantitative. For SOLIS with a fixed branching factor $W=3$, the maximum nodes searched at depth S is $N(S) = \sum_{i=0}^S 3^i$, giving $N(3) = 40$, $N(4) = 121$, and $N(5) = 364$. Stockfish, by contrast, explores a variable number of nodes per move depending on pruning and heuristics: at depth 3 it averaged 137 nodes (maximum 1625), at depth 4 214 nodes (maximum 4463), and at depth 5 338 nodes (maximum 10611). SOLIS thus matches this Stockfish configuration while examining an order of magnitude fewer nodes at the higher depths. The conclusion is the central empirical message of the paper: **much of the strength normally gained from broad expansion can be recovered from higher-quality representations alone.**

4.3 INTERPRETABLE GAME TRAJECTORIES

Our fourth contribution is interpretability, and this subsection makes it concrete by tracing latent sequences of moves from real games. Figure 5 shows three representative examples—a game won by White, a balanced draw, and a game won by Black—each plotted over the same 2D projection from a sample dataset, with independently embedded positions connected by arrows to show the progression of play. The reading is intuitive: decisive games follow smooth paths through the space, while closely contested games fluctuate around the center. These trajectories complement the embedding visualization of Figure 1 by showing that planning is not a black box—it corresponds to visible, directed movement toward winning or losing regions, exactly as our advantage-vector formulation predicts.

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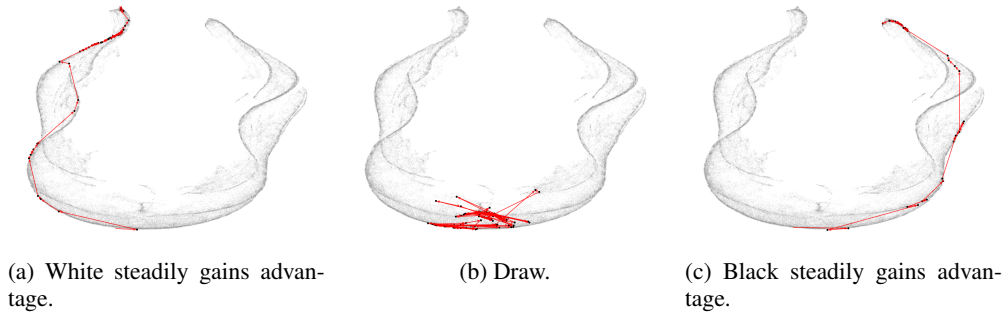


Figure 5: Latent trajectory visualizations supporting our interpretability contribution: three games embedded in the shared representation space, with red arrows tracing how positions progress as play unfolds. Decisive games move steadily toward a winning region; the draw hovers near the neutral center.

5 DISCUSSION

5.1 LIMITATIONS

The very design choices that give SOLIS its contributions—a history-free representation and a search that ranks moves solely by advantage alignment—also bound what it can do. We surface two limitations that follow directly from these choices, both illustrated by concrete games.

Blindness to threefold repetition SOLIS selects moves by alignment with the advantage vector, which yields effective decisions at minimal overhead but discards move history. The most consequential side effect is blindness to threefold repetition: if the same position (including side to move, castling, and en-passant rights) recurs three times, either player can claim a draw. A savvy opponent can engineer repeats—e.g., perpetual checks—to neutralize winning positions SOLIS would otherwise convert. Game C in Appendix A.4 shows SOLIS reaching a winning position against Stockfish yet conceding a premature draw by repetition. This limitation is a direct trade-off against the simplicity that powers our efficiency contribution, and points to rule-aware state features as future work.

Indecisiveness in winning positions Because evaluation alignment is our planning signal, SOLIS cannot distinguish forcing positions of different lengths: both mate-in-1 and mate-in-20 carry a win probability of 1.0, though the shorter line should be preferred. Choosing the longer continuation leaves room for the opponent to complicate matters, occasionally yielding unnecessary draws or even losses. Ruoss et al. (2024) met the same issue and sidestepped it by consulting Stockfish as an external oracle; SOLIS instead insists on its own solution, which remains a gap to close. Game I in Appendix A.5 shows SOLIS holding many winning continuations for several moves in a row yet drifting into a draw. Like the previous point, this limitation is the flip side of a deliberate design decision, and preferring shorter forced wins is a natural extension.

5.2 CONCLUSION

This paper set out to test a single claim—that planning can be carried out in an evaluation-aligned latent space, with no policy, value head, or dynamics model—and delivered four contributions toward it. *First*, we showed that supervised contrastive learning yields an embedding in which proximity reflects outcome similarity and a single global *advantage vector* points from losing to winning regions. *Second*, we reduced planning in that space to vector arithmetic, ranking candidate actions by alignment with the advantage direction under two scoring rules and a small search. *Third*, we demonstrated that this suffices for competitive chess. *Fourth*, we showed that the resulting play is interpretable as movement through the learned space.

The chess evidence for these claims is concrete. Using a fixed branching factor $W=3$ and shallow depths $S \in \{3, 4, 5\}$, SOLIS matches or exceeds a time-capped Stockfish in several equal-depth matchups while searching far fewer nodes per move (40–364 for SOLIS versus hundreds to thou-

sands for Stockfish at the same depths). Elo rises with depth; with anchored scoring both Mini and Base exceed 2500 Elo at depth five, and anchoring adds a consistent 10–30 Elo for Base. Width brings little benefit past three candidates—the recurring signal that strength stems from the quality of the learned space rather than broad expansion. The embedding also yields readable game paths, with decisive games moving steadily toward winning regions and draws hovering near neutral.

Taken together, these contributions establish evaluation-aligned latent planning as a simple and compute-efficient alternative to rollout-based search or explicit dynamics models. We expect the same idea to extend to other perfect-information games and to control tasks with large action spaces, where low-branching search is valuable. Building on the limitations above, future work includes adding rule-aware state features, preferring shorter forced wins, combining SOLIS with policy priors or MCTS, studying calibration and uncertainty, and testing transfer beyond chess.

REPRODUCIBILITY STATEMENT

Each contribution is accompanied by the material needed to reproduce it. The full training setup, including hyperparameters and implementation details, is provided in Appendix A.2. Section 3 describes the encoder architecture, input representation, supervised contrastive objective (our first contribution), and inference procedure (our second), with model configurations summarized in Table 1. Evaluation details supporting the strength contribution—match setup, Stockfish configurations, and rating calculations—are given in Section 4 and Appendix A.3, and detailed match results are included in Appendix A.6 for Elo verification.

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A APPENDIX

A.1 USE OF LARGE LANGUAGE MODELS (LLMs)

Large language models were used only for minor tasks such as formatting and figure/table placement. They were not used to design experiments, analyze results, or generate technical content, and every methodological and experimental contribution reported here was developed and verified solely by the authors.

A.2 TRAINING SETUP

To support reproduction of our first contribution, we record the full training recipe. Both SOLIS configurations (Mini, Base) are trained with supervised contrastive learning on 5M ChessBench positions, using cosine similarity with ℓ_2 -normalized embeddings and SGD with momentum on six 48 GB L40S GPUs for 400k steps (batch size 128).

The two most consequential settings are τ and δ , since together they shape the evaluation-aligned geometry. The temperature $\tau = 0.07$ follows prior contrastive work (Radford et al., 2021; Girdhar et al., 2023), where smaller values sharpen similarity scores and larger ones oversmooth gradients; sweeping $\tau \in \{0.05, 0.07, 0.1\}$, we found 0.07 converged most reliably across model sizes. The margin $\delta = 0.05$ balances collapsing too many positions into one neighborhood (large δ) against fragmenting the space (small δ), and also reflects Stockfish evaluation noise, since sub-0.05 differences are rarely meaningful in practice.

A.3 STOCKFISH SETUP

For fair and reproducible comparison, all experiments use the official Linux release of Stockfish 16 (Romstad et al., 2008) with a 32 MB hash and default threading. Rating caps are set with its `UCI_LimitStrength` and `UCI_Elo` options, and depths with the `depth` parameter. The nodes-searched figures underpinning our efficiency contribution are obtained by setting the engine information level to `INFO_ALL` and extracting the `nodes` attribute for each move.

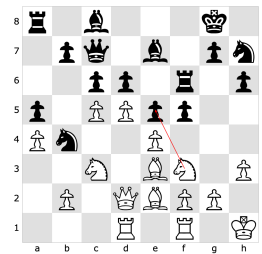
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Table 3: SOLIS training hyperparameters.

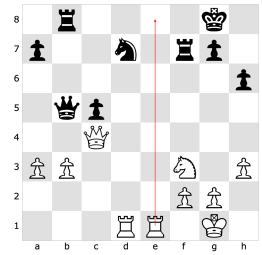
Attribute	Value
Loss	SupCon
τ	0.07
δ (positive margin)	0.05
Positives per anchor	5
Optimizer	SGD
Momentum	0.9
Learning rate	0.05
Batch size	128
Steps	400k
Dropout	0.10
Activation	GELU
Similarity	cosine
Embedding norm	ℓ_2
Token length	77

A.4 REPRESENTATIVE GAMES: STOCKFISH (CAP) VS. SOLIS (DEPTH, WIDTH)

To make the strength and interpretability contributions tangible at the level of individual play, we provide sample games between SOLIS and Stockfish, with SOLIS limited to search depth S and width W and Stockfish given unbounded search under a rating cap. Games *A*, *B*, and *E* showcase SOLIS' tactical understanding and long-term planning, while games *C* and *D* illustrate the two limitations discussed in Section 5. In each table, the left portion encodes the moves in Portable Game Notation (PGN) and the right portion displays a critical moment.

<pre> 1. d4 Nf6 2. c4 d6 3. Nc3 e5 4. Nf3 Qe7 5. e4 c6 6. d5 Qc7 7. Be2 h6 8. a4 a5 9. O-O Be7 10. h3 Na6 11. Be3 Nb4 12. Kh1 Nh7 13. Qd2 O-O 14. Rad1 f5 15. c5 Rf6 16. Nxe5 f4 17. Bxf4 Rxf4 18. Qxf4 dxe5 19. d6 exf4 20. Bc4+ Kf8 21. dxc7 Ke8 22. e5 Ng5 23. Rd6 Bxd6 24. exd6 Bd7 25. Re1+ Kf8 26. Re7 Bf5 27. h4 Nd5 28. Nxd5 cxd5 29. Bxd5 Be6 30. Bxe6 f3 31. hxg5 b6 32. Bh3 hxg5 33. Re5 Rc8 34. Rf5+ Kg8 35. Rf7 g4 36. cxb6 g6 37. d7 fxg2+ 38. Kh2 Rf8 39. Bxg2 Kxf7 40. Bc6 Ke7 41. Kg3 Rf3+ 42. Kh2 Rd3 43. b7 Rh3+ 44. Kg2 Re3 45. d8=Q+ Kf7 46. b8=Q Rg3+ 47. Kh2 Rh3+ 48. Kg1 Rh7 49. Qe8+ Kf6 50. Qbd8+ Kf5 51. Be4+ Kf4 52. Qf6# 1-0 </pre>	
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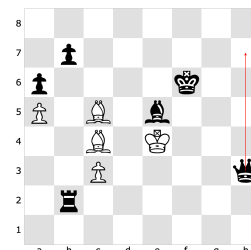
Game A — SOLIS ($S=5$, $W=3$) vs. SF 2350, SOLIS–White. Result: 1–0. Illustrating the strength contribution, SOLIS sacrifices a knight to seize a middlegame advantage.

<pre> 1. d4 d5 2. e3 c6 3. Bd3 h6 4. h3 Nf6 5. Nf3 e6 6. O-O b6 7. c4 Ba6 8. b3 Nbd7 9. Bb2 Bb4 10. a3 Be7 11. Nbd2 O-O 12. Re1 c5 13. Qe2 Qc7 14. e4 Nxe4 15. Nxe4 dxe4 16. Bxe4 Rab8 17. d5 Bf6 18. dxe6 Bxb2 19. exf7+ Rxf7 20. Qxb2 Bb7 21. Bxb7 Qxb7 22. Rad1 b5 23. Qe2 bxc4 24. Qxc4 Qb5 25. Re8+ Nf8 26. Rxb8 Qxb8 27. Rd5 Qf4 28. Qxf4 Rxf4 29. Rxc5 Ne6 30. Rc6 Nd4 31. Nxd4 a5 32. Rc4 Rf8 33. Nc6 Kh8 34. Nxa5 Ra8 35. b4 Rd8 36. g3 Rd3 37. a4 g5 38. b5 Rd2 39. b6 Rd5 40. b7 Rxa5 41. Rc8+ Kh7 42. b8=Q Rd5 43. Rc7+ Kg6 44. Qg8+ Kf5 45. Qxd5+ Kg6 46. Qf7# 1-0 </pre>	
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Game B — SOLIS ($S=5$, $W=3$) vs. SF 2300, SOLIS–White. Result: 1–0. Further evidence of SOLIS' tactical strength: it sacrifices a rook to win Stockfish's queen and converts the advantage into a win.

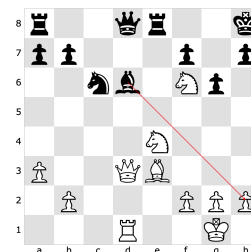
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1. e3 Nf6 2. Nc3 d5 3. Nf3 c5 4. d4 e6 5. g3 a6 6. a4 Nc6 7. Ne2 Bd6 8. Bg2 O-O 9. h3 Re8 10. c3 Qc7 11. Nd2 e5 12. Nb3 c4 13. dxe5 Nxe5 14. Nbd4 Nd3+ 15. Kf1 Nc5 16. Bf3 Bd7 17. Kg2 Be5 18. a5 Rad8 19. Qc2 g6 20. Bd2 h5 21. Be1 h4 22. gxh4 Kg7 23. Ng3 Rh8 24. h5 Rdg8 25. hxxg6 fxg6 26. Be2 Qd6 27. Qd1 Qe7 28. f3 Bd6 29. Nf1 Rh5 30. h4 Rgh8 31. Bf2 R5h7 32. Qe1 Bb8 33. Bg3 Ba7 34. Bf2 Nh5 35. Ng3 Bb8 36. f4 Nf6 37. h5 Rh6 38. hxxg6 Rxh1 39. Qxxh1 Qe8 40. Ndf5+ Bxf5 41. Nxf5+ Kxg6 42. Ne7+ Kf7 43. e4 Rxh1 44. Rxh1 Qxe7 45. Bxc5 Qxe4+ 46. Bf3 Qg6+ 47. Kf1 Bxf4 48. b4 cxb3 49. Rh8 Ke6 50. Rf8 b2 51. Rxf6+ Kxf6 52. Bxd5 b1=R+ 53. Ke2 Rb2+ 54. Kf3 Be5 55. Bc4 Qh5+ 56. Ke3 Qh3+ 57. Ke4 Qh7+ 58. Kf3 Qh3+ 59. Ke4 Qh7+ 60. Kf3 Qh3+ 61. Ke4 Qh7+ 62. Kf3 Qh3+ 63. Ke4 Qh7+ 64. Kf3 Qh3+ 65. Ke4 1/2-1/2



Game C — SF 2200 vs. SOLIS ($S=5$, $W=3$), SOLIS-Black. Result: 1/2-1/2. A concrete instance of the first limitation: SOLIS reaches a winning endgame yet settles for a draw because of its blindness to threefold repetition.

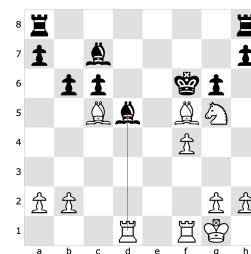
1. d4 d5 2. c4 dxc4 3. e3 e5 4. Bxc4 exd4 5. exd4 Nf6 6. Nc3 Bd6 7. Qe2+ Qe7 8. Be3 O-O 9. Nf3 c6 10. O-O Be6 11. Rfd1 Re8 12. d5 cxd5 13. Bxd5 Nc6 14. a3 Nxd5 15. Nxd5 Qd8 16. Qd3 Bg4 17. Ng5 g6 18. Ne4 Bxd1 19. Ndf6+ Kh8 20. Rxd1 Bxh2+ 21. Kxh2 Qxd3 22. Rxd3 Red8 23. Nd7 Kg7 24. f3 h6 25. Rd6 f5 26. Nec5 Re8 27. Bd2 Re2 28. Kh1 Kf7 29. b3 Ke7 30. Nxb7 Rc8 31. Ndc5 Kf7 32. Kg1 Kg8 33. Rxg6+ Kf7 34. Rd6 Ne7 35. Rd7 Rg8 36. Nd6+ Kg6 37. Nc4 Kh5 38. Ne3 f4 39. Ne6 fxe3 40. Nf4+ Kg5 41. Nxe2 exd2 42. Kf2 Kf6 43. Nc3 Ke6 44. Rd4 Rc8 45. Ne4 d1=N+ 46. Rxd1 Rc2+ 47. Kg1 Nf5 48. Rd2 Rxd2 49. Nxd2 Ke5 50. Kh2 Kf4 51. b4 Ke3 52. Ne4 Nd4 53. a4 Nc2 54. b5 Nd4 55. Nd6 Ne6 56. a5 Kd3 57. g4 Kd4 58. b6 axb6 59. axb6 Nc5 60. Kg3 Kd5 61. b7 Na6 62. Nf7 Kc6 63. Nxb6 Kxb7 64. f4 Kc7 65. Kh4 Kd7 66. Kh5 Nc5 67. f5 Kd6 68. f6 Ne6 69. Nf7+ Kc6 70. Kh6 Kd5 71. g5 Nc5 72. Kg6 Ne6 73. Kh5 Kd4 74. g6 Kc4 75. Ne5+ Kd5 76. Ng4 Ke4 77. Kh6 Kf5 78. g7 Nxg7 79. Kxg7 Kxg4 80. Kf7 Kf5 81. Ke7 Kg5 82. f7 Kf4 83. Kf6 Kf3 84. Ke5 Kg3 85. Ke4 Kf2 86. Kd3 Kg1 87. Ke3 Kh2 88. f8=Q Kg3 89. Qg8+ Kh3 90. Kf4 Kh2 91. Kf3 Kh3 92. Qg3# 1-0



Game D — SF 2500 vs. SOLIS ($S=5$, $W=3$), SOLIS-Black. Result: 1-0 A cautionary case for the strength contribution: SOLIS makes a short-sighted piece sacrifice that leads to a lasting material disadvantage.

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1. d4 d5 2. e3 Nd7 3. c4 e5 4. dxe5 dxc4 5. Nf3
Qe7 6. Bxc4 Nxe5 7. Nxe5 Qxe5 8. Qb3 Qh5 9. e4
Nf6 10. O-O Bd6 11. Qb5+ c6 12. Qxh5 Nxh5 13. Nc3
Be5 14. f4 Bc7 15. Be3 g6 16. Bb3 Be6 17. Rad1 Ng7
18. Bc2 f5 19. Bc5 Kf7 20. exf5 Nxf5 21. Ne4 b6
22. Ng5+ Kf6 23. Bxf5 Bd5 24. Rxd5 cxd5 25. Bd4+
Kxf5 26. h3 Bxf4 27. Bxh8 Kxg5 28. Bg7 Rg8 29. Bd4
Re8 30. Rf3 Re1+ 31. Kf2 Re6 32. g3 Bd2 33. Rf7
Re4 34. Bf6+ Kh6 35. g4 a6 36. Kg3 d4 37. h4 Be1+
38. Kh3 Re3+ 39. Kg2 Bxh4 40. Bxh4 a5 41. Bf6 a4
42. Rd7 Rd3 43. Rd5 b5 44. Rxb5 a3 45. Re5 Rd2+
46. Kf3 axb2 47. g5+ Kh5 48. Rb5 Rd3+ 49. Kf4 h6
50. gxh6+ Kxh6 51. Rxb2 g5+ 52. Kg4 Rd1 53. Bxg5+
Kg6 54. Rb6+ Kg7 55. Kf5 Kf7 56. Rb7+ Ke8 57. a4
Ra1 58. Re7+ Kf8 59. Rh7 Rxa4 60. Bh6+ Kg8 61. Re7
Rc4 62. Kg5 Rc5+ 63. Kg4 Ra5 64. Rg7+ Kh8 65. Re7
d3 66. Rd7 d2 67. Bxd2 Rc5 68. Bf4 Kg8 69. Be3 Rb5
70. Bh6 Re5 71. Kf4 Rc5 72. Ke4 Rb5 73. Kf4 Rb4+
74. Ke5 Rb1 75. Kf6 Rb2 76. Kg6 Rb6+ 77. Kh5 Rb2
78. Bg5 Rb6 79. Bh6 Re6 80. Kg5 Ra6 81. Rg7+ Kh8
82. Re7 Ra5+ 83. Kg6 Ra6+ 84. Kh5 Kg8 85. Rb7 Rc6
86. Rd7 Rc5+ 87. Bg5 Rc2 88. Kg6 Rc6+ 89. Bf6 Re6
90. Rd8+ Re8 91. Rxe8# 1-0

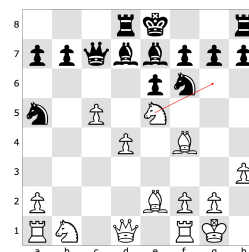


Game E — SOLIS ($S=5, W=3$) vs. Stockfish 2500 , SOLIS–White. Result: 1-0 Reinforcing the strength contribution, SOLIS makes an exchange sacrifice for long-term compensation and converts the advantage into a win.

A.5 EQUAL-DEPTH MATCHES: SOLIS *D* vs. STOCKFISH *D*

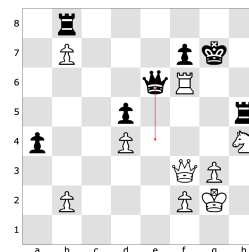
These games reinforce the efficiency contribution at the board level: SOLIS and Stockfish search to the same depth *D*, with no rating cap on Stockfish, so any edge SOLIS shows comes from its latent planning rather than a handicap. As before, the left portion of each table encodes the moves in Portable Game Notation (PGN) and the right portion displays a critical moment.

1. d4 d5 2. e3 c5 3. Nf3 Nf6 4. Be2 e6 5. O-O Nc6 6. b3 cxd4 7. exd4 Be7 8. c3 Qc7 9. h3 Bd7 10. c4 dxc4 11. bxc4 Na5 12. Ne5 Rd8 13. Bf4 Bd6 14. c5 Be7 15. Ng6 Qc8 16. Nxf8 Bc6 17. Nc3 Kf8 18. Rc1 Qa8 19. Bc7 b5 20. Bxa5 Bxg2 21. d5 Bxh3 22. Qd3 Bf5 23. Qg3 b4 24. Rfd1 Bxc5 25. Qc7 Be7 26. Nb5 Nxd5 27. Rxd5 Qxd5 28. Bxb4 Qd7 29. Qxd7 Re8 30. Rc8 g5 31. Qxe7+ Kg7 32. Qxf7+ Kh6 33. Bf8+ Rxf8 34. Qxf8# 1-0



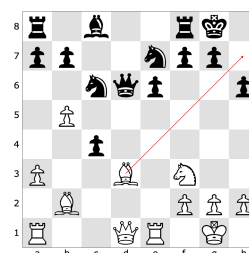
Game F — SOLIS $S = 5$ vs. Stockfish $S = 5$, SOLIS-White. Result: 1-0 At equal depth, SOLIS sacrifices a knight in the middlegame for a concrete tactical advantage—strength without broad search.

1. d4 d5 2. e3 c5 3. Nf3 e6 4. Be2 cxd4 5. exd4 Bd6 6. O-O Nf6 7. Bg5 Nbd7 8. Nbd2 h6 9. Bh4 O-O 10. Re1 a5 11. Bf1 Bf4 12. c3 Qb6 13. Qc2 Qc7 14. a4 b6 15. Bb5 Ba6 16. Bxa6 Rxa6 17. Qd3 Ra7 18. Re2 Qc8 19. Ree1 Nh5 20. Nf1 Nh6 21. N1d2 Nh5 22. Nf1 Bc7 23. Qd1 Nh6 24. Qd3 Nh5 25. Qd1 Nh6 26. Qd3 Rb7 27. Qb5 Ra7 28. N1d2 Nh5 29. h3 g5 30. Rac1 gxf4 31. c4 Nf4 32. Rc3 Qd8 33. g3 Nxf4+ 34. Kg2 Ng5 35. cxd5 h3+ 36. Kg1 h2+ 37. Kxh2 exd5 38. Rc6 Nf6 39. Nxf5 hxf5 40. Kg2 Re8 41. Rh1 Bd6 42. Nf3 Kg7 43. Nxf5 Rc7 44. Rxd6 Qxd6 45. Qf1 b5 46. axb5 Rcc8 47. Rh4 Rh8 48. Nf3 Ne4 49. Rf4 Rh5 50. Qe2 Qe6 51. Nh4 Rch8 52. b6 Nf6 53. Qf3 a4 54. b7 Rb8 55. Rxf6 Qe4 56. Rxf7+ Kh8 57. Qxe4 dxe4 58. Rc7 Rg5 59. Rc8+ Rg8 60. Rc7 Rg5 61. Rc8+ Rg8 62. Ng6+ Kg7 63. Ne7 Rf8 64. Rc7 Rf7 65. d5 Rbf8 66. Nf5+ Kg6 67. Rxf7 Kxf7 68. Nd6+ Kg6 69. Nc8 Kg5 70. b8=Q Kg4 71. Qc7 Rf5 72. Nd6 e3 73. Qg7+ Rg5 74. f3+ Kh5 75. Qh7# 1-0



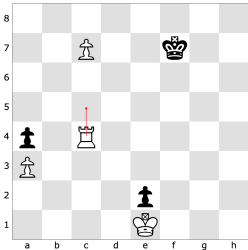
Game G — SOLIS $S = 5$ vs. Stockfish $S = 5$, SOLIS-White. Result: 1-0 At equal depth, Stockfish steadily gains an advantage, then blunders in the late middlegame; SOLIS capitalizes on the error and wins the endgame.

1. d4 c6 2. e4 d5 3. Nc3 Nf6 4. e5 Ng8 5. Bd3 e6 6. Nf3 c5 7. dxc5 Bxc5 8. O-O Nc6 9. a3 h6 10. b4 Bb6 11. Nb5 Bc7 12. Re1 Bb8 13. c4 Nge7 14. Nd6+ Bxd6 15. exd6 Qxd6 16. Bb2 O-O 17. b5 dxc4 18. Bh7+ Kh8 19. Qxd6 f6 20. Be4 c3 21. Bxc3 e5 22. bxc6 Bf5 23. Qxe7 Bxe4 24. Rxe4 bxc6 25. Nxe5 Rfe8 26. Nf7+ Kg8 27. Nxf6+ Kh7 28. Rae1 Rad8 29. Ba5 Rd4 30. g4 Rxe4 31. Rxe4 Rb8 32. Qxa7 Rd8 33. Rf4 Rd3 34. g5 fxf5 35. Rd4 Rxa3 36. Kf1 Rb3 37. Rb4 Rc3 38. h4 gxf4 39. Rd4 h3 40. Qe7 h2 41. Qe4+ Kxh6 42. Rd6+ Kg5 43. Rg6+ Kh5 44. Qg4# 1-0



Game H — SOLIS $S = 3$ vs. Stockfish $S = 3$, SOLIS-White. Result: 1-0 At equal depth, SOLIS sacrifices a bishop for a discovered attack on Stockfish's queen and converts the advantage into a win.

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1. d4 d5 2. e3 c6 3. c4 e6 4. Nf3 Nf6 5. Be2 c5 6. O-O Nc6 7. b3 a6 8. Bb2 Be7 9. a3 cxd4 10. exd4 O-O 11. c5 b6 12. b4 bxc5 13. dxc5 Bb7 14. Nbd2 a5 15. b5 Bxc5 16. Rc1 Qb6 17. bxc6 Ng4 18. Rxc5 Ba6 19. Rc2 Nxf2 20. Rxf2 Bxe2 21. Qxe2 a4 22. c7 Rac8 23. Bd4 Qb5 24. Qxb5 Rfe8 25. Qb8 h6 26. Ne5 g5 27. h4 Rf8 28. Ne4 dxe4 29. Rxf7 Rxb8 30. Rxf8+ Rxf8 31. hxc5 Rc8 32. gxh6 e3 33. Bxe3 Kh7 34. Rc4 Kg8 35. Kh2 Kh7 36. g4 Rg8 37. g5 Ra8 38. Rg4 Kh8 39. Ba7 Rf8 40. Bb8 Rf2+ 41. Kg1 Rc2 42. Rc4 Rb2 43. Ba7 Rb1+ 44. Kg2 Kh7 45. Bf2 Rb2 46. g6+ Kxh6 47. Kh2 Rxf2+ 48. Kg3 Rg2+ 49. Kxg2 Kg7 50. Rc1 Kf6 51. Kh2 Kg7 52. Nd7 e5 53. Nf6 Kxf6 54. Rg1 Kg7 55. Rf1 Kh6 56. Rg1 Kg7 57. Rf1 Kxg6 58. Rc1 Kg7 59. Rc4 Kf7 60. Kg2 Ke7 61. Kf2 Kd6 62. Ke1 Kd5 63. Rc1 Ke4 64. Ke2 Kd4 65. Rc2 e4 66. Rc1 Kd5 67. Ke1 Ke5 68. Rc4 Kd5 69. Rc2 Kd6 70. Rc1 Ke7 71. Rc4 Kf7 72. Rc5 e3 73. Kf1 Ke7 74. Rc4 Kf7 75. Kg2 e2 76. Kf2 Ke7 77. Ke1 Kf7 78. Rc5 Kg7 79. Rc2 Kf6 80. Rc4 Kf7 81. Rc5 Kg6 82. Kd2 Kf7 83. Ke1 Ke7 84. Rc4 Kf7 85. Rc5 Kg6 86. Rc4 Kf7 87. Rc5 1/2-1/2	
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Game I — SOLIS $S = 3$ vs. Stockfish $S = 3$, SOLIS–White. Result: 1/2-1/2 A concrete instance of the second limitation: SOLIS struggles to plan in a position where every trajectory is winning, and settles for a draw.

A.6 SOLIS VS. STOCKFISH RESULTS

To make the strength contribution fully reproducible, we report the complete match outcomes behind our Elo estimates, with SOLIS searching at depths $S \in \{2, 3, 4, 5\}$. Each table is indexed by Stockfish’s Elo cap (rows) and SOLIS search width W (columns); entries give win–loss–draw counts with total match points in parentheses. We provide separate tables for the Base and Mini models and for both the unanchored and anchored scoring rules.

Base: Depth 2					Base: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	32–46–22 (55)	41–47–12 (64)	–	–	1500	–	–	–	–
1550	39–38–23 (58)	37–50–13 (62)	–	–	1550	–	–	–	–
1600	32–44–24 (54)	33–51–16 (58)	–	–	1600	–	–	–	–
1650	33–41–26 (54)	29–46–25 (52)	–	–	1650	–	–	–	–
1700	34–38–28 (53)	22–53–25 (48)	–	–	1700	–	–	–	–
1750	37–35–28 (54)	30–49–21 (54)	–	–	1750	–	–	–	–
1800	29–42–29 (50)	36–42–22 (57)	–	–	1800	39–29–32 (54)	37–49–14 (62)	–	–
1850	–	–	–	–	1850	35–32–33 (51)	30–50–20 (55)	–	–
1900	–	–	–	–	1900	37–30–33 (52)	40–41–19 (60)	–	–
1950	–	–	–	–	1950	31–28–41 (45)	37–44–19 (59)	–	–
2000	–	–	34–40–26 (54)	43–31–26 (58)	2000	41–27–32 (54)	33–45–22 (56)	–	–
2050	–	–	46–38–16 (65)	36–38–26 (55)	2050	39–27–34 (52)	31–41–28 (52)	–	–
2100	–	–	34–45–21 (56)	34–31–35 (50)	2100	41–27–32 (54)	36–38–26 (55)	–	–
2150	–	–	46–31–23 (62)	30–43–27 (52)	2150	–	–	–	–
2200	–	–	31–35–34 (48)	35–28–37 (49)	2200	–	–	33–42–25 (54)	30–41–29 (50)
2250	–	–	34–34–32 (51)	40–29–31 (54)	2250	–	–	35–41–24 (56)	33–43–24 (54)
2300	–	–	32–32–36 (48)	37–34–29 (54)	2300	–	–	35–33–32 (52)	36–34–30 (53)
2350	–	–	–	–	2350	–	–	37–37–26 (56)	29–37–34 (48)
2400	–	–	–	–	2400	–	–	34–36–30 (52)	37–38–25 (56)
2450	–	–	–	–	2450	–	–	37–28–35 (51)	43–29–28 (58)
2500	–	–	–	–	2500	–	–	41–30–29 (56)	34–30–36 (49)
2550	–	–	–	–	2550	–	–	–	–
2600	–	–	–	–	2600	–	–	–	–
2650	–	–	–	–	2650	–	–	–	–
2700	–	–	–	–	2700	–	–	–	–

Base: Depth 4					Base: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	–	–	–	–	1600	–	–	–	–
1650	–	–	–	–	1650	–	–	–	–
1700	–	–	–	–	1700	–	–	–	–
1750	–	–	–	–	1750	–	–	–	–
1800	–	–	–	–	1800	–	–	–	–
1850	–	–	–	–	1850	–	–	–	–
1900	–	–	–	–	1900	–	–	–	–
1950	37–26–37 (50)	41–42–17 (62)	–	–	1950	36–34–30 (53)	38–41–21 (58)	–	–
2000	43–20–37 (53)	31–43–26 (52)	–	–	2000	31–28–41 (45)	27–52–21 (53)	–	–
2050	38–27–35 (52)	40–38–22 (59)	–	–	2050	41–24–35 (53)	36–37–27 (54)	–	–
2100	42–25–33 (54)	32–48–20 (56)	–	–	2100	43–23–34 (54)	36–39–25 (56)	–	–
2150	36–23–41 (48)	31–47–22 (54)	–	–	2150	42–23–35 (54)	27–45–28 (50)	–	–
2200	46–13–41 (52)	39–39–22 (58)	–	–	2200	42–21–37 (52)	23–51–26 (48)	–	–
2250	–	–	–	–	2250	–	–	–	–
2300	–	–	–	–	2300	–	–	–	–
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2450	–	–	40–39–21 (60)	39–38–23 (58)	2450	–	–	25–46–29 (48)	43–30–27 (58)
2500	–	–	38–38–24 (57)	41–40–19 (61)	2500	–	–	40–36–24 (58)	34–44–22 (56)
2550	–	–	44–20–36 (54)	42–29–29 (56)	2550	–	–	40–31–29 (56)	38–38–24 (57)
2600	–	–	46–17–37 (54)	37–23–40 (48)	2600	–	–	46–25–29 (58)	33–37–30 (52)
2650	–	–	41–18–41 (50)	44–22–34 (55)	2650	–	–	47–17–36 (56)	30–35–35 (48)
2700	–	–	45–16–39 (53)	42–22–36 (53)	2700	–	–	38–25–37 (50)	45–21–34 (56)

Figure 6: Full match record behind our strength contribution—Base model vs. Stockfish, per-depth blocks under unanchored planning. Entries are W–D–L (score) out of 100.

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Mini: Depth 2					Mini: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	32–32–36 (48)	27–45–28 (50)	–	–	1350	–	–	–	–
1400	30–41–29 (50)	33–27–40 (46)	–	–	1400	–	–	–	–
1450	29–44–27 (51)	33–40–27 (53)	–	–	1450	–	–	–	–
1500	31–38–31 (50)	31–39–30 (50)	–	–	1500	29–38–33 (48)	28–43–29 (50)	–	–
1550	38–35–27 (56)	23–50–27 (48)	–	–	1550	41–36–23 (59)	36–37–27 (54)	–	–
1600	34–30–36 (49)	29–37–34 (48)	–	–	1600	37–30–33 (52)	37–30–33 (52)	–	–
1650	24–45–31 (46)	28–36–36 (46)	–	–	1650	32–38–30 (51)	30–40–30 (50)	–	–
1700	–	–	–	–	1700	28–37–35 (46)	21–43–36 (42)	–	–
1750	–	–	–	–	1750	36–33–31 (52)	28–40–32 (48)	–	–
1800	–	–	37–31–32 (52)	37–36–27 (55)	1800	39–24–37 (51)	29–35–36 (46)	–	–
1850	–	–	34–24–42 (46)	34–31–35 (50)	1850	–	–	–	–
1900	–	–	40–28–32 (54)	40–29–31 (54)	1900	–	–	–	–
1950	–	–	39–29–32 (54)	39–29–32 (54)	1950	–	–	–	–
2000	–	–	38–34–28 (55)	42–28–30 (56)	2000	–	–	34–35–31 (52)	38–21–41 (48)
2050	–	–	41–27–32 (54)	40–26–34 (53)	2050	–	–	38–34–28 (55)	34–33–33 (50)
2100	–	–	35–30–35 (50)	36–26–38 (49)	2100	–	–	33–34–33 (50)	34–29–37 (48)
2150	–	–	–	–	2150	–	–	31–35–34 (48)	38–25–37 (50)
2200	–	–	–	–	2200	–	–	38–18–44 (47)	31–26–43 (44)
2250	–	–	–	–	2250	–	–	36–23–41 (48)	45–26–29 (58)
2300	–	–	–	–	2300	–	–	40–22–38 (51)	34–26–40 (47)
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2500	–	–	–	–	2500	–	–	–	–
Mini: Depth 4					Mini: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	–	–	–	–	1350	–	–	–	–
1400	–	–	–	–	1400	–	–	–	–
1450	–	–	–	–	1450	–	–	–	–
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	36–42–22 (57)	35–31–34 (50)	–	–	1600	–	–	–	–
1650	29–40–31 (49)	36–33–31 (52)	–	–	1650	–	–	–	–
1700	33–34–33 (50)	36–32–32 (52)	–	–	1700	39–30–31 (54)	34–34–32 (51)	–	–
1750	43–26–31 (56)	28–38–34 (47)	–	–	1750	37–29–34 (52)	32–30–38 (47)	–	–
1800	38–24–38 (50)	30–37–33 (48)	–	–	1800	42–20–38 (52)	33–31–36 (48)	–	–
1850	49–23–28 (60)	29–33–38 (46)	–	–	1850	36–24–40 (48)	43–28–29 (57)	–	–
1900	–	–	–	–	1900	45–26–29 (58)	46–31–23 (62)	–	–
1950	–	–	–	–	1950	40–22–38 (51)	37–35–28 (54)	–	–
2000	–	–	–	–	2000	–	–	–	–
2050	–	–	–	–	2050	–	–	–	–
2100	–	–	–	–	2100	–	–	–	–
2150	–	–	31–31–38 (46)	44–21–35 (54)	2150	–	–	–	–
2200	–	–	35–29–36 (50)	48–27–25 (62)	2200	–	–	27–37–36 (46)	33–38–29 (52)
2250	–	–	37–31–32 (52)	37–28–35 (51)	2250	–	–	30–33–37 (46)	38–31–31 (54)
2300	–	–	27–37–36 (46)	36–34–30 (53)	2300	–	–	38–37–25 (56)	33–39–28 (52)
2350	–	–	22–42–36 (43)	37–35–28 (54)	2350	–	–	32–29–39 (46)	29–36–35 (47)
2400	–	–	36–29–35 (50)	37–26–37 (50)	2400	–	–	30–26–44 (43)	30–32–38 (46)
2500	–	–	–	–	2500	–	–	26–34–40 (43)	43–25–32 (56)

Figure 7: Full match record behind our strength contribution—Mini model vs. Stockfish, per-depth blocks under unanchored planning. Entries are W–D–L (score) out of 100.

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Base: Depth 2					Base: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	32–46–22 (55)	41–47–12 (64)	–	–	1500	–	–	–	–
1550	39–38–23 (58)	37–50–13 (62)	–	–	1550	–	–	–	–
1600	32–44–24 (54)	33–51–16 (58)	–	–	1600	–	–	–	–
1650	33–41–26 (54)	29–46–25 (52)	–	–	1650	–	–	–	–
1700	34–38–28 (53)	22–53–25 (48)	–	–	1700	–	–	–	–
1750	37–35–28 (54)	30–49–21 (54)	–	–	1750	–	–	–	–
1800	29–42–29 (50)	36–42–22 (57)	–	–	1800	39–29–32 (54)	37–49–14 (62)	–	–
1850	–	–	–	–	1850	35–32–33 (51)	30–50–20 (55)	–	–
1900	–	–	–	–	1900	37–30–33 (52)	40–41–19 (60)	–	–
1950	–	–	–	–	1950	31–28–41 (45)	37–44–19 (59)	–	–
2000	–	–	34–40–26 (54)	43–31–26 (58)	2000	41–27–32 (54)	33–45–22 (56)	–	–
2050	–	–	46–38–16 (65)	36–38–26 (55)	2050	39–27–34 (52)	31–41–28 (52)	–	–
2100	–	–	34–45–21 (56)	34–31–35 (50)	2100	41–27–32 (54)	36–38–26 (55)	–	–
2150	–	–	46–31–23 (62)	30–43–27 (52)	2150	–	–	–	–
2200	–	–	31–35–34 (48)	35–28–37 (49)	2200	–	–	33–42–25 (54)	30–41–29 (50)
2250	–	–	34–34–32 (51)	40–29–31 (54)	2250	–	–	35–41–24 (56)	33–43–24 (54)
2300	–	–	32–32–36 (48)	37–34–29 (54)	2300	–	–	35–33–32 (52)	36–34–30 (53)
2350	–	–	–	–	2350	–	–	37–37–26 (56)	29–37–34 (48)
2400	–	–	–	–	2400	–	–	34–36–30 (52)	37–38–25 (56)
2450	–	–	–	–	2450	–	–	37–28–35 (51)	43–29–28 (58)
2500	–	–	–	–	2500	–	–	41–30–29 (56)	34–30–36 (49)
2550	–	–	–	–	2550	–	–	–	–
2600	–	–	–	–	2600	–	–	–	–
2650	–	–	–	–	2650	–	–	–	–
2700	–	–	–	–	2700	–	–	–	–
Base: Depth 4					Base: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	–	–	–	–	1600	–	–	–	–
1650	–	–	–	–	1650	–	–	–	–
1700	–	–	–	–	1700	–	–	–	–
1750	–	–	–	–	1750	–	–	–	–
1800	–	–	–	–	1800	–	–	–	–
1850	–	–	–	–	1850	–	–	–	–
1900	–	–	–	–	1900	–	–	–	–
1950	37–26–37 (50)	41–42–17 (62)	–	–	1950	36–34–30 (53)	38–41–21 (58)	–	–
2000	43–20–37 (53)	31–43–26 (52)	–	–	2000	31–28–41 (45)	27–52–21 (53)	–	–
2050	38–27–35 (52)	40–38–22 (59)	–	–	2050	41–24–35 (53)	36–37–27 (54)	–	–
2100	42–25–33 (54)	32–48–20 (56)	–	–	2100	43–23–34 (54)	36–39–25 (56)	–	–
2150	36–23–41 (48)	31–47–22 (54)	–	–	2150	42–23–35 (54)	27–45–28 (50)	–	–
2200	46–13–41 (52)	39–39–22 (58)	–	–	2200	42–21–37 (52)	23–51–26 (48)	–	–
2250	–	–	–	–	2250	–	–	–	–
2300	–	–	–	–	2300	–	–	–	–
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2450	–	–	40–39–21 (60)	39–38–23 (58)	2450	–	–	25–46–29 (48)	43–30–27 (58)
2500	–	–	38–38–24 (57)	41–40–19 (61)	2500	–	–	40–36–24 (58)	34–44–22 (56)
2550	–	–	44–20–36 (54)	42–29–29 (56)	2550	–	–	40–31–29 (56)	38–38–24 (57)
2600	–	–	46–17–37 (54)	37–23–40 (48)	2600	–	–	46–25–29 (58)	33–37–30 (52)
2650	–	–	41–18–41 (50)	44–22–34 (55)	2650	–	–	47–17–36 (56)	30–35–35 (48)
2700	–	–	45–16–39 (53)	42–22–36 (53)	2700	–	–	38–25–37 (50)	45–21–34 (56)

Figure 8: Full match record behind our strength contribution—Base model vs. Stockfish, per-depth blocks under anchored planning. Entries are W–D–L (score) out of 100.

1188
1189
1190
1191
1192
1193
1194
1195
1196
1197
1198
1199
1200
1201
1202
1203
1204
1205
1206
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1210
1211
1212
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1215
1216
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1229
1230
1231
1232
1233
1234
1235
1236
1237
1238
1239
1240
1241

Mini: Depth 2					Mini: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	36–32–32 (52)	39–30–31 (54)	–	–	1350	–	–	–	–
1400	23–49–28 (48)	30–32–38 (46)	–	–	1400	–	–	–	–
1450	33–37–30 (52)	24–47–29 (48)	–	–	1450	–	–	–	–
1500	34–38–28 (53)	25–40–35 (45)	–	–	1500	29–40–31 (49)	35–33–32 (52)	–	–
1550	28–38–34 (47)	40–39–21 (60)	–	–	1550	34–35–31 (52)	37–29–34 (52)	–	–
1600	23–47–30 (46)	28–42–30 (49)	–	–	1600	32–36–32 (50)	32–39–29 (52)	–	–
1650	35–29–36 (50)	38–37–25 (56)	–	–	1650	34–30–36 (49)	37–33–30 (54)	–	–
1700	–	–	–	–	1700	36–30–34 (51)	34–35–31 (52)	–	–
1750	–	–	–	–	1750	34–25–41 (46)	34–30–36 (49)	–	–
1800	–	–	35–37–28 (54)	40–28–32 (54)	1800	32–29–39 (46)	32–31–37 (48)	–	–
1850	–	–	32–32–36 (48)	44–29–27 (58)	1850	–	–	–	–
1900	–	–	37–30–33 (52)	45–21–34 (56)	1900	–	–	–	–
1950	–	–	43–30–27 (58)	36–29–35 (50)	1950	–	–	–	–
2000	–	–	35–31–34 (50)	43–27–30 (56)	2000	–	–	46–32–22 (62)	38–27–35 (52)
2050	–	–	37–29–34 (52)	45–27–28 (58)	2050	–	–	37–27–36 (50)	39–26–35 (52)
2100	–	–	36–27–37 (50)	42–34–24 (59)	2100	–	–	29–34–37 (46)	48–17–35 (56)
2150	–	–	–	–	2150	–	–	37–33–30 (54)	39–25–36 (52)
2200	–	–	–	–	2200	–	–	37–31–32 (52)	41–23–36 (52)
2250	–	–	–	–	2250	–	–	36–23–41 (48)	44–20–36 (54)
2300	–	–	–	–	2300	–	–	30–43–27 (52)	38–26–36 (51)
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2500	–	–	–	–	2500	–	–	–	–
Mini: Depth 4					Mini: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	–	–	–	–	1350	–	–	–	–
1400	–	–	–	–	1400	–	–	–	–
1450	–	–	–	–	1450	–	–	–	–
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	26–45–29 (48)	31–33–36 (48)	–	–	1600	–	–	–	–
1650	37–39–24 (56)	28–32–40 (44)	–	–	1650	–	–	–	–
1700	23–39–38 (42)	29–41–30 (50)	–	–	1700	37–36–27 (55)	34–29–37 (48)	–	–
1750	36–29–35 (50)	38–30–32 (53)	–	–	1750	42–23–35 (54)	37–33–30 (54)	–	–
1800	35–24–41 (47)	33–36–31 (51)	–	–	1800	36–28–36 (50)	31–35–34 (48)	–	–
1850	41–21–38 (52)	30–28–42 (44)	–	–	1850	38–21–41 (48)	33–32–35 (49)	–	–
1900	–	–	–	–	1900	37–27–36 (50)	37–37–26 (56)	–	–
1950	–	–	–	–	1950	43–18–39 (52)	38–24–38 (50)	–	–
2000	–	–	–	–	2000	–	–	–	–
2050	–	–	–	–	2050	–	–	–	–
2100	–	–	–	–	2100	–	–	–	–
2150	–	–	39–23–38 (50)	32–30–38 (47)	2150	–	–	–	–
2200	–	–	33–40–27 (53)	38–25–37 (50)	2200	–	–	30–40–30 (50)	32–34–34 (49)
2250	–	–	36–31–33 (52)	30–27–43 (44)	2250	–	–	29–43–28 (50)	49–26–25 (62)
2300	–	–	32–33–35 (48)	37–28–35 (51)	2300	–	–	31–30–39 (46)	30–35–35 (48)
2350	–	–	35–34–31 (52)	40–25–35 (52)	2350	–	–	26–32–42 (42)	40–34–26 (57)
2400	–	–	28–26–46 (41)	26–38–36 (45)	2400	–	–	39–32–29 (55)	31–32–37 (47)
2500	–	–	–	–	2500	–	–	30–29–41 (44)	28–41–31 (48)

Figure 9: Full match record behind our strength contribution—Mini model vs. Stockfish, per-depth blocks under anchored planning. Entries are W–D–L (score) out of 100.